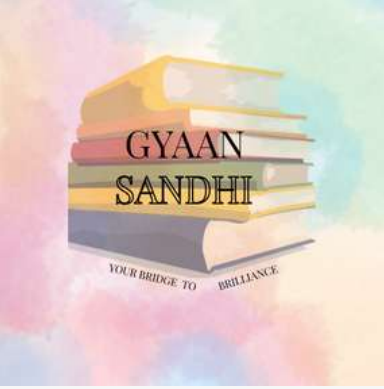




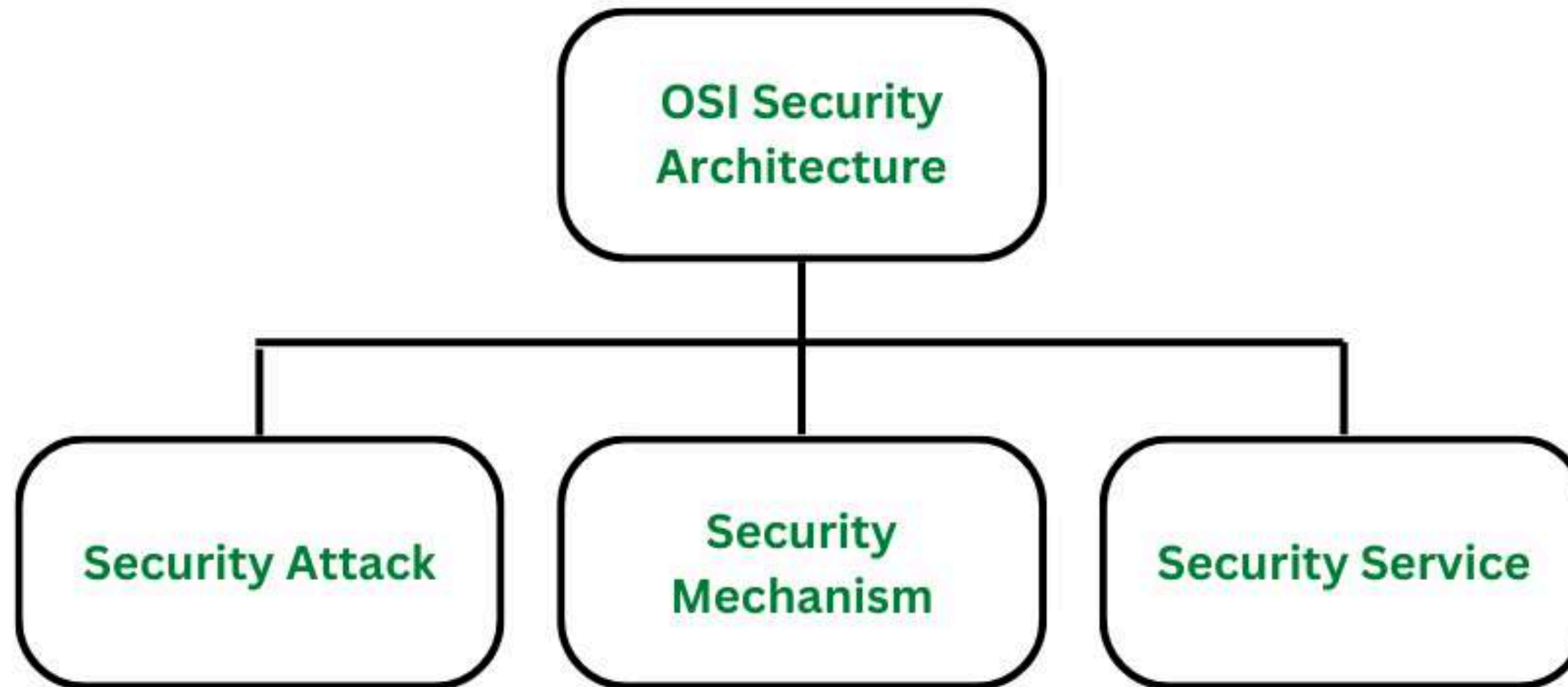
Information Security

Unit 5

**Network and System
Security**

OSI Security Architecture

- OSI Security Architecture is a framework developed by the International Organization for Standardization to define security attacks, security services, and security mechanisms used to protect data in network communication.
- It provides a structured way to understand how security can be implemented in computer networks.



1) Security Attacks

A security attack is any action that tries to compromise the security of information.

Types of Security Attacks

A) Passive Attack

In passive attack, the attacker only monitors or reads data without changing it.

Examples:

- Eavesdropping
- Reading confidential messages
- Traffic analysis

Characteristics:

- Difficult to detect
- Data is not modified

B) Active Attack

In active attack, the attacker modifies or disrupts data.

Examples:

- Data modification
- Masquerade attack
- Replay attack
- Denial of Service (DoS)

Characteristics:

- Easier to detect
- System/data gets affected

2) Security Services

Security services are techniques provided to protect data and communication.

Types of Security Services

A) Authentication

Verifies the identity of users or systems.

Example:

- Login using username and password

B) Access Control

Prevents unauthorized users from accessing resources.

Example:

- Admin-only files

C) Data Confidentiality

Protects data from unauthorized access.

Example:

- Encryption

D) Data Integrity

Ensures data is not modified during transmission.

Example:

- Hashing

E) Non-Repudiation

Prevents sender or receiver from denying communication.

Example:

- Digital signatures

F) Availability

Ensures services and data are available when needed.

Example:

- Backup servers

3) Security Mechanisms

Security mechanisms are methods used to implement security services.

Types of Security Mechanisms

Specific Security Mechanisms

- Encryption
- Digital Signature
- Access Control
- Authentication Exchange
- Traffic Padding

Pervasive Security Mechanisms

- Trusted Functionality
- Security Labels
- Event Detection
- Security Audit Trail

Security Mechanism	Meaning
Encryption	Converts normal data into unreadable form (cipher text) to protect it from unauthorized access.
Digital Signature	Used to verify sender identity and ensure data integrity and non-repudiation.
Access Control	Restricts unauthorized users from accessing systems or resources.
Authentication Exchange	Process of verifying identity between communicating users or systems.
Traffic Padding	Adds extra dummy data into communication to confuse attackers and prevent traffic analysis.
Trusted Functionality	Ensures that security-related functions of a system work correctly and reliably.
Security Labels	Tags assigned to data/resources indicating their security level or classification.
Event Detection	Detects security-related events such as attacks or unauthorized activities.
Security Audit Trail	Maintains logs/records of system activities for monitoring and investigation purposes.

Access Control

- Access Control is a security mechanism that restricts unauthorized users from accessing systems, networks, data, or resources.
- It ensures that only authorized users can access specific information or perform certain actions.

Objectives of Access Control

- Protect confidential data
- Prevent unauthorized access
- Ensure system security
- Maintain data integrity
- Limit user permissions

Working of Access Control

- Access control works in three steps:
- Identification
- User identifies themselves using username or ID.
- Authentication
- System verifies identity using password, OTP, fingerprint, etc.
- Authorization
- System checks what resources or actions the user is allowed to access.

Access Control

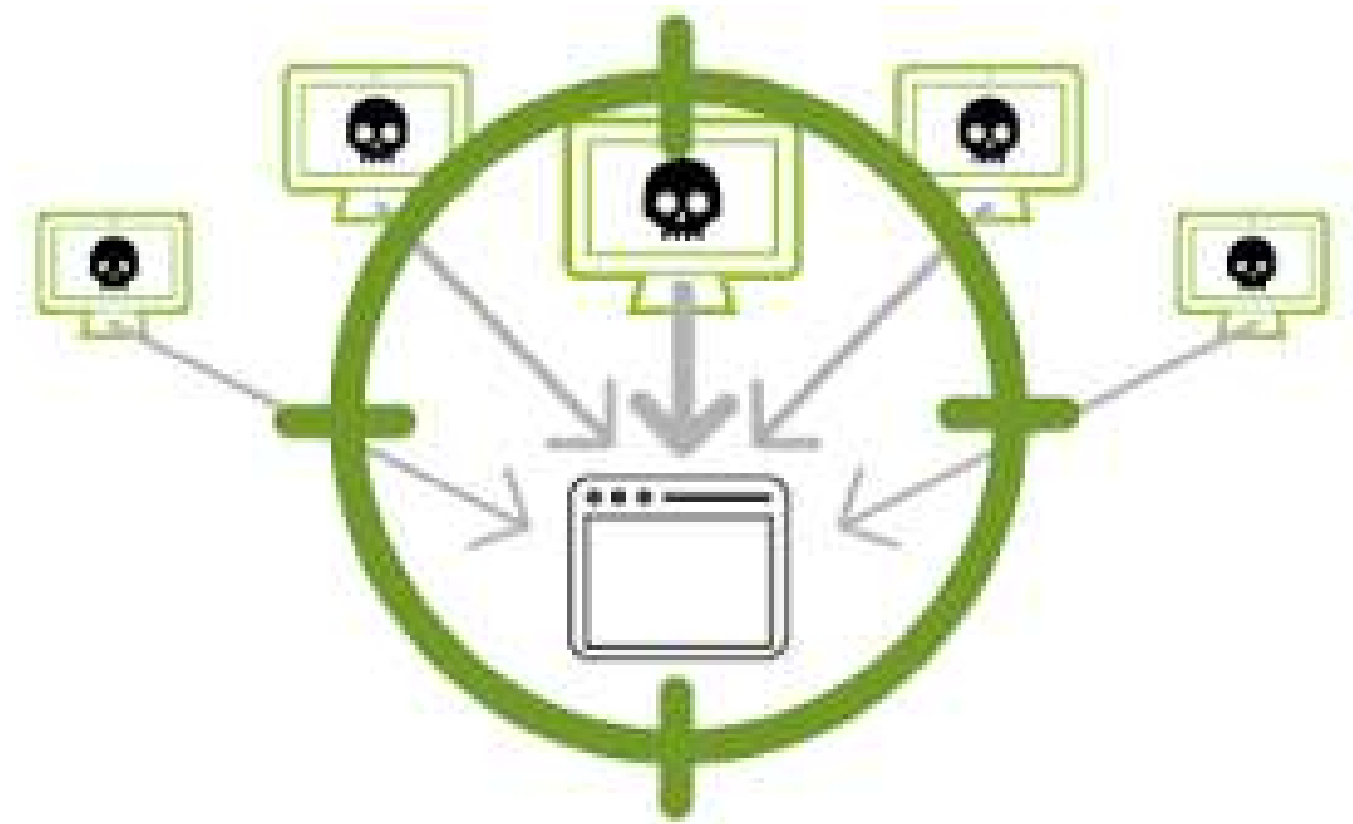
Type	Full Form	Description	Example	Advantages	Disadvantages
DAC	Discretionary Access Control	Resource owner decides who can access data	File owner gives read/write permission	Flexible and easy to manage	Less secure
MAC	Mandatory Access Control	Central authority controls access based on security levels	Military systems	Highly secure	Less flexible
RBAC	Role-Based Access Control	Access provided according to user roles	Admin, Teacher, Student permissions	Easy management	Complex role setup
ABAC	Attribute-Based Access Control	Access based on attributes like time, location, device	Access only during office hours	Fine-grained security	Complex implementation

Flooding Attacks

- A Flooding Attack is a type of cyber attack in which an attacker sends a huge amount of traffic, requests, or data to a network, server, or system to overload it and make the service unavailable to legitimate users.
- Flooding attacks are commonly used in Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks.

Working of Flooding Attack

1. Attacker sends massive traffic to the target system.
2. System resources such as bandwidth, memory, or CPU become overloaded.
3. Legitimate users cannot access the service.
4. The server may slow down or crash completely.



Flooding Attacks

Type of Attack	Description
SYN Flood Attack	Attacker sends many fake TCP connection requests to exhaust server resources
ICMP Flood Attack	Huge number of ICMP ping packets are sent to the target
UDP Flood Attack	Large amount of UDP packets are transmitted to overload the system
HTTP Flood Attack	Excessive HTTP requests are sent to crash web servers
Smurf Attack	ICMP packets with spoofed IP address flood the victim system

Prevention Methods

- Firewalls
- Intrusion Detection Systems (IDS)
- Traffic filtering
- Rate limiting
- Load balancing
- Anti-DDoS protection systems

Denial of Service (DoS) Attack

- A Denial of Service (DoS) attack is a cyber attack in which an attacker attempts to make a computer system, server, or network unavailable to legitimate users by overwhelming it with excessive traffic or requests.
- The main goal of a DoS attack is to disrupt normal service.

Working of DoS Attack

1. Attacker sends a large number of requests or packets to the target server.
2. Server resources such as CPU, memory, and bandwidth become overloaded.
3. The system becomes slow or crashes.
4. Genuine users cannot access the service.

Characteristics of DoS Attack

- Performed by a single system or attacker
- Targets server availability
- Causes resource exhaustion
- Disrupts normal network operations

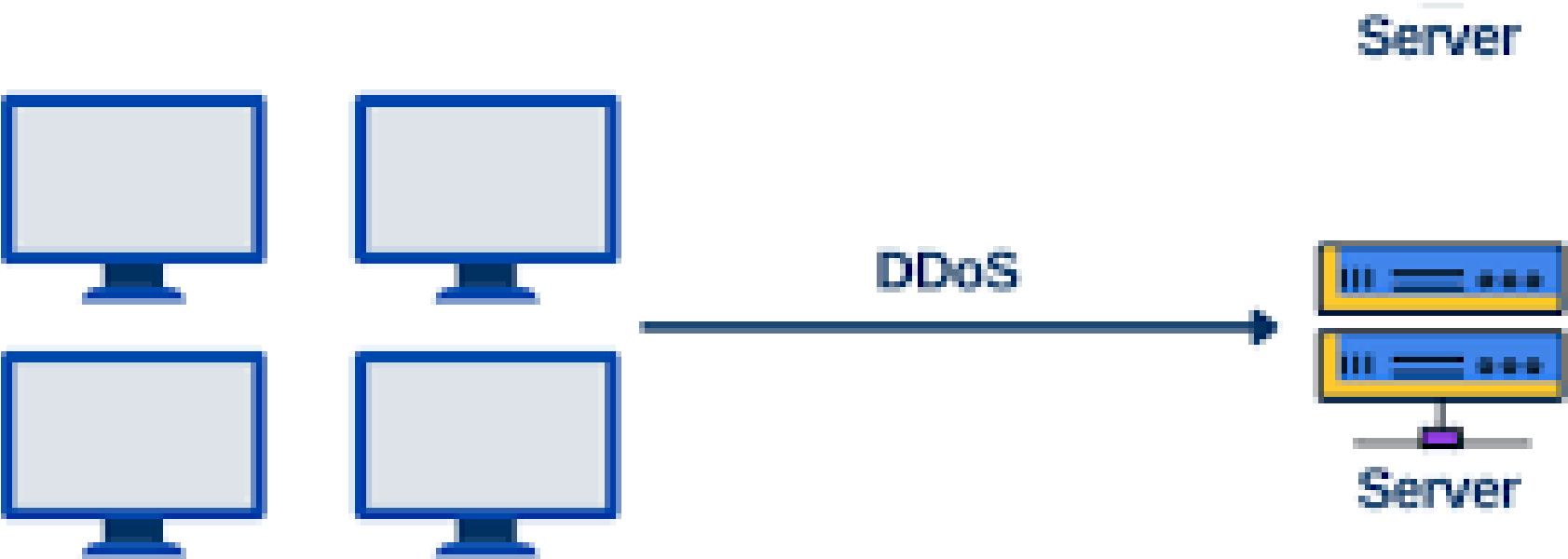


Distributed Denial of Service (DDoS) Attack

- A Distributed Denial of Service (DDoS) attack is a cyber attack in which multiple compromised systems are used to send massive traffic to a target server, network, or website to make it unavailable to legitimate users.
- Unlike a DoS attack, DDoS attacks are launched from many systems simultaneously.

Working of DDoS Attack

- Attacker infects multiple computers with malware.
- These infected systems form a botnet.
- Botnet sends huge traffic to the target server.
- Server resources become overloaded.
- Legitimate users cannot access the service.



Type	Description
Volume-Based Attack	Floods bandwidth with huge traffic
Protocol Attack	Exploits network protocol weaknesses
Application Layer Attack	Targets web applications using HTTP requests

Feature	DoS (Denial of Service)	DDoS (Distributed Denial of Service)
Full Form	Denial of Service	Distributed Denial of Service
Number of Attack Systems	Single system used for attack	Multiple systems used for attack
Attack Source	One attacker machine	Many infected devices (botnet)
Traffic Volume	Smaller traffic	Very large traffic
Detection	Easier to detect	Difficult to detect
Attack Strength	Less powerful	More powerful
Speed of Attack	Comparatively slower	Faster and massive
Complexity	Simple to perform	Complex to organize
Impact	Temporary service disruption	Large-scale service failure
Example	Single computer flooding a server	Thousands of bots attacking a website simultaneously

Intrusion Detection

- Intrusion Detection is the process of monitoring a computer system or network to identify unauthorized access, malicious activities, or security policy violations.
- The system used for this purpose is called an Intrusion Detection System (IDS).

Objectives of Intrusion Detection

- Detect unauthorized access
- Identify malicious activities
- Protect network resources
- Generate security alerts
- Improve system security

Working of IDS

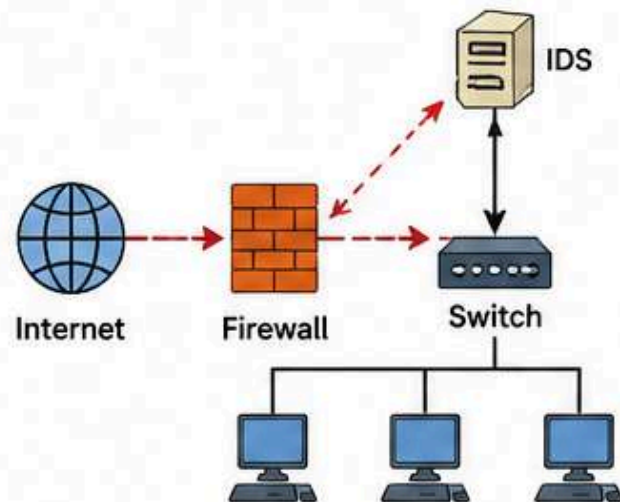
- 1.IDS monitors network traffic or system activities.
- 2.It analyzes packets and events.
- 3.Suspicious behavior is detected.
- 4.Alerts are generated for administrators.

Type	Description
Network-Based IDS (NIDS)	Monitors entire network traffic
Host-Based IDS (HIDS)	Monitors activities on a specific device
Signature-Based IDS	Detects known attack patterns
Anomaly-Based IDS	Detects unusual behavior from normal activity

TYPES OF INTRUSION DETECTION SYSTEMS (IDS)

IDS monitors network or system activities to detect unauthorized access, attacks or policy violations.

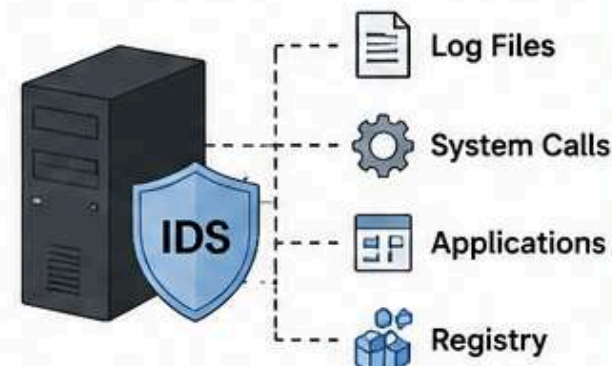
1. NETWORK-BASED IDS (NIDS)



Monitors the entire network traffic and analyzes packets to detect suspicious activities.

Used for:
Large networks, gateways,
routers, firewalls.

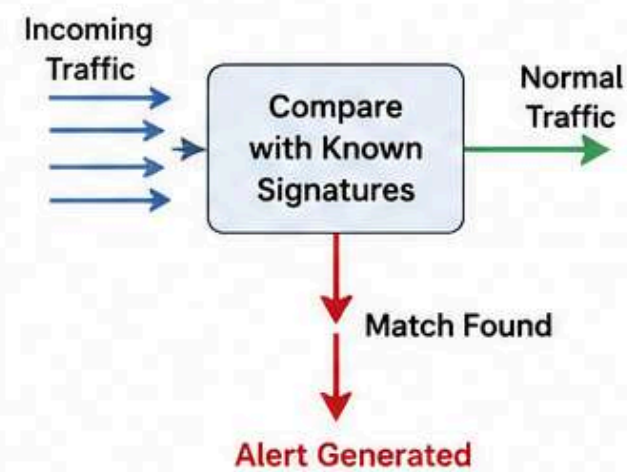
2. HOST-BASED IDS (HIDS)



Monitors activities on a specific host or device such as log files, system calls, and file integrity.

Used for:
Servers, endpoints, databases,
workstations.

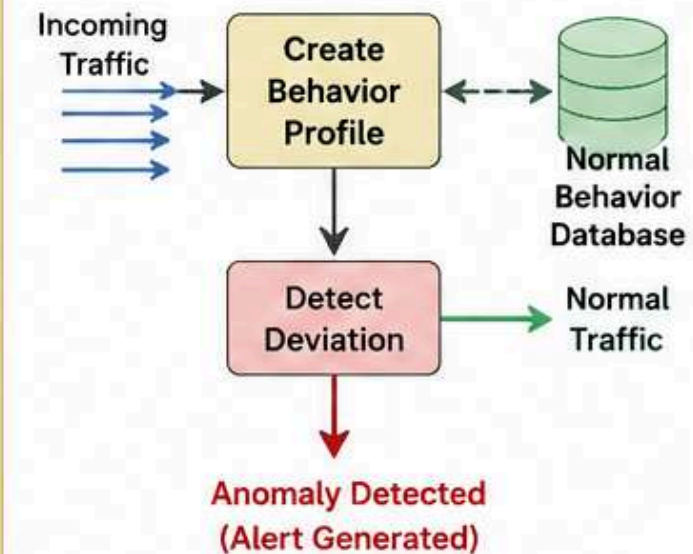
3. SIGNATURE-BASED IDS (MISUSE DETECTION)



Detects attacks by comparing activities with known attack patterns or signatures.

Used for:
Known threats and attacks
with existing signatures.

4. ANOMALY-BASED IDS (BEHAVIOR ANALYSIS)



Detects attacks by identifying unusual or abnormal behavior from normal activity.

Used for:
Unknown threats, zero-day
attacks, insider attacks.



Note: IDS can be used alone or in combination to provide strong security for networks and systems.

Feature	Anomaly-Based IDS	Signature-Based IDS
Definition	Detects attacks by identifying abnormal behavior	Detects attacks using known attack patterns/signatures
Detection Method	Compares current behavior with normal behavior	Matches traffic with stored attack signatures
Unknown Attack Detection	Can detect new or unknown attacks	Cannot detect unknown attacks
Accuracy	May generate false alarms	More accurate for known attacks
Database Requirement	Does not depend heavily on signature database	Requires regular signature updates
Speed	Slower due to behavior analysis	Faster detection process
Example	Detecting unusual network traffic	Detecting known malware pattern

Host-Based Honeypot

A Host-Based Honeypot is a security system installed on a single computer or host to attract attackers and monitor malicious activities on that specific device.

It behaves like a real system to trap hackers and study their actions.

Features of Host-Based Honeypot

- Installed on a single machine
- Monitors local activities
- Collects attack information
- Detects unauthorized access
- Simulates real operating systems or applications

Working of Host-Based Honeypot

- Honeypot is installed on a host system.
- Attacker tries to access or attack the system.
- Honeypot records attacker activities.
- Security administrators analyze the collected data.

Network-Based Honeypot

A Network-Based Honeypot is a security mechanism placed within a network to monitor and analyze attacks targeting the entire network infrastructure.

It simulates vulnerable network services and devices to attract attackers.

Features of Network-Based Honeypot

- Monitors network traffic
- Detects external attacks
- Simulates multiple systems/services
- Collects network attack data
- Protects large networks

Working of Network-Based Honeypot

- Honeypot is connected within the network.
- Fake vulnerable services are exposed.
- Attackers attempt to exploit the services.
- Honeypot captures and logs attack traffic.
- Security teams analyze attack patterns.

Firewall

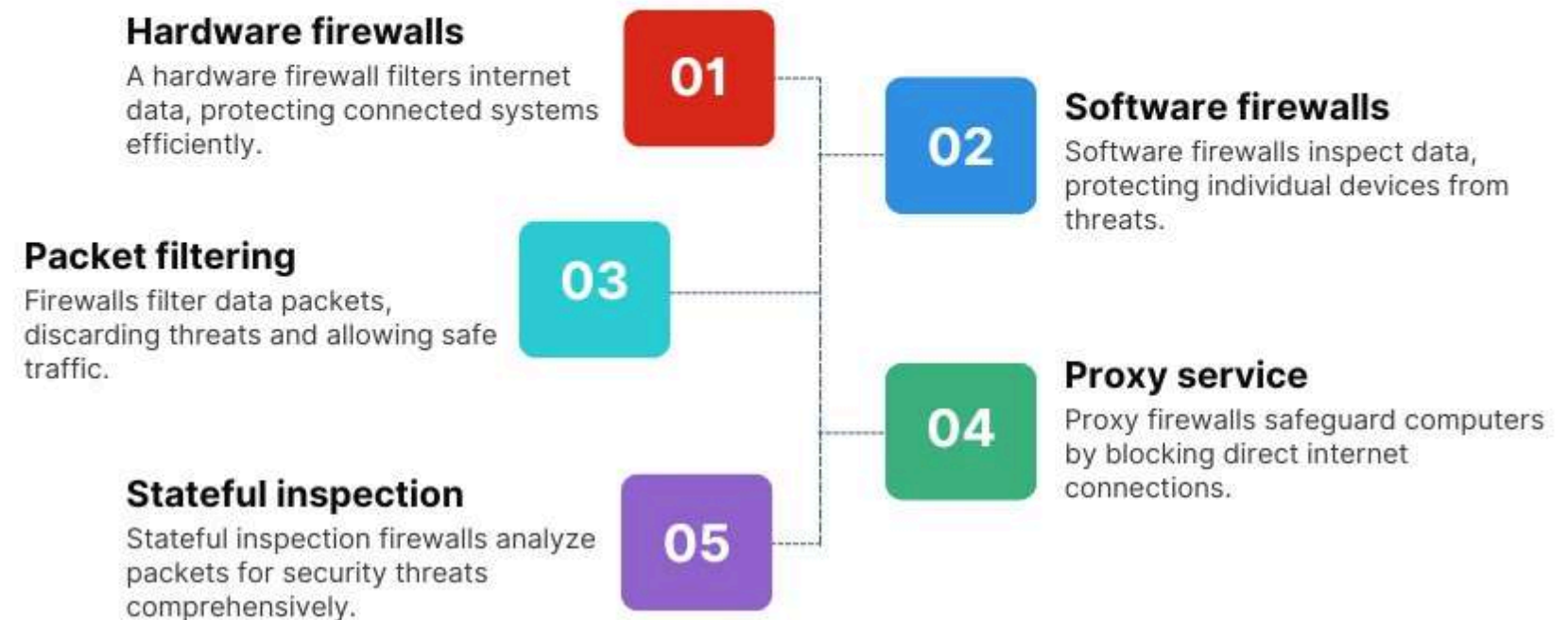
A Firewall is a network security system that monitors and controls incoming and outgoing network traffic based on predefined security rules.

It acts as a barrier between a trusted internal network and an untrusted external network such as the internet.

Working of Firewall

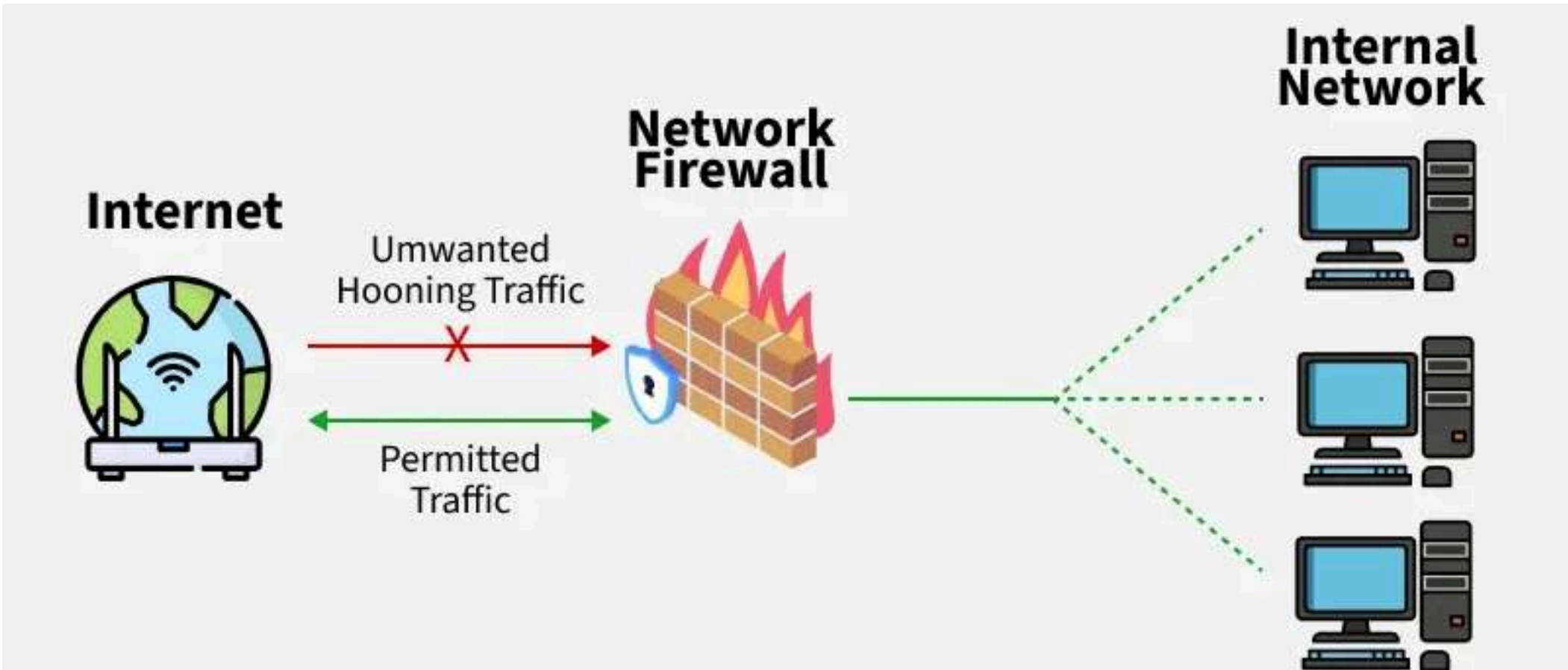
- 1.Data packets enter the network.
- 2.Firewall checks packets using security rules.
- 3.Allowed traffic passes through.
- 4.Suspicious or unauthorized traffic is blocked.

Types of Firewall



Firewall

Type	Description
Packet Filtering Firewall	Filters packets based on IP address and port number
Stateful Inspection Firewall	Monitors active connections and packet states
Proxy Firewall	Acts as intermediary between user and internet
Application Firewall	Filters traffic at application layer
Next-Generation Firewall (NGFW)	Advanced firewall with IDS/IPS and deep inspection



Firewall

Need	Explanation
Prevent Unauthorized Access	Blocks hackers and unauthorized users
Network Security	Protects systems from cyber attacks
Traffic Monitoring	Monitors incoming and outgoing traffic
Malware Protection	Helps stop malicious traffic
Data Protection	Secures confidential information
Access Control	Restricts access to specific resources

Firewall Characteristics

Characteristic	Description
Single Entry Point	All traffic passes through firewall
Security Policy Enforcement	Applies predefined security rules
Traffic Monitoring	Continuously monitors network packets
Authentication Support	Verifies users and devices
Logging and Auditing	Maintains records of activities

Intrusion Prevention System (IPS)

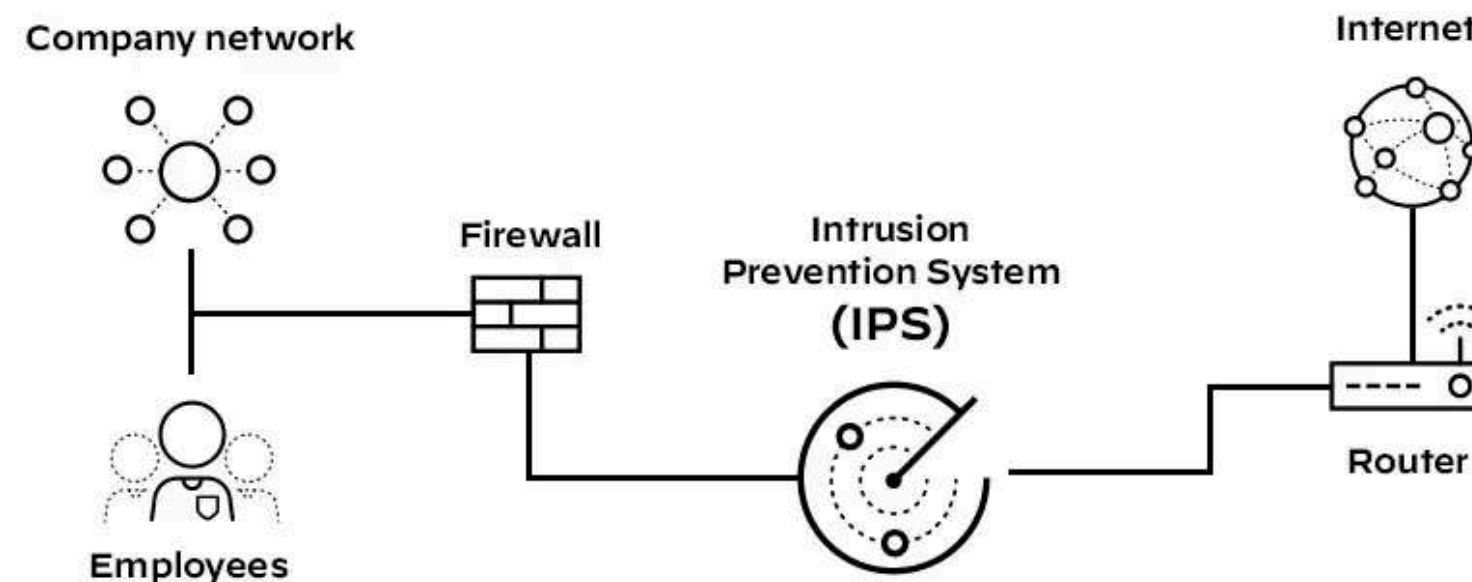
An Intrusion Prevention System (IPS) is a network security technology that monitors network traffic to detect and automatically prevent malicious activities or attacks.

Unlike IDS, IPS can actively block attacks in real time.

Working of IPS

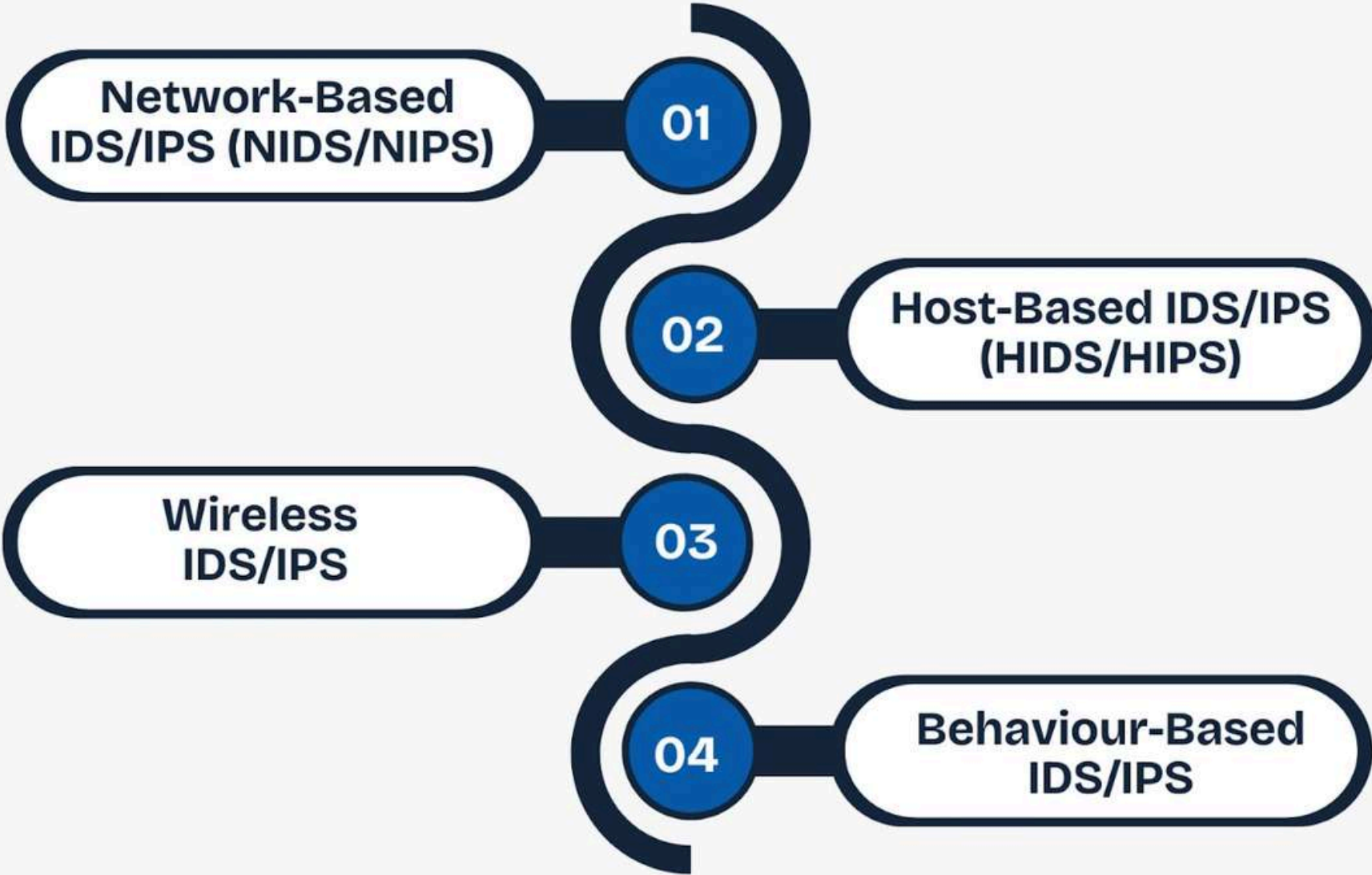
- IPS monitors network traffic continuously.
- Suspicious packets or attacks are identified.
- IPS blocks or drops malicious traffic automatically.
- Alerts are generated for administrators.

Intrusion Prevention Systems



Types of IPS

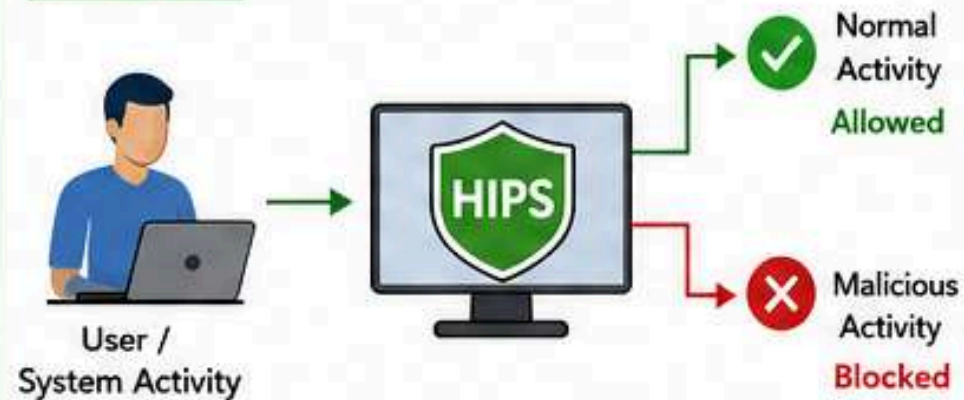
Type	Description
Network-Based IPS (NIPS)	Monitors entire network traffic
Host-Based IPS (HIPS)	Protects individual host systems
Wireless IPS (WIPS)	Protects wireless networks
Network Behavior Analysis IPS	Detects abnormal traffic behavior



1) HOST-BASED IPS (HIPS)

Installed on individual host devices to monitor and prevent malicious activities inside that system.

DIAGRAM



FEATURES

- Protects individual devices
- Monitors system files and applications
- Detects malware and suspicious behavior
- Blocks unauthorized access

ADVANTAGES

- Strong endpoint protection
- Detects insider attacks
- Protects critical systems

DISADVANTAGES

- Uses system resources
- Limited to one host

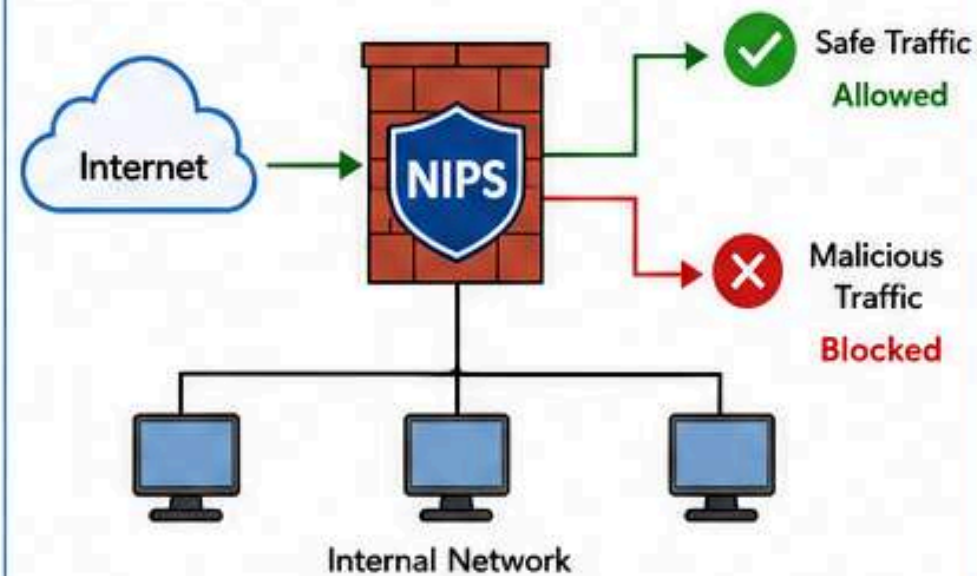
EXAMPLE

Protecting a server or PC

2) NETWORK-BASED IPS (NIPS)

Monitors and protects entire network traffic to detect and prevent attacks across the network.

DIAGRAM



FEATURES

- Monitors network packets
- Detects network attacks
- Blocks malicious traffic
- Provides centralized security

ADVANTAGES

- Protects entire network
- Real-time attack prevention
- Centralized monitoring

DISADVANTAGES

- Expensive setup
- Complex management

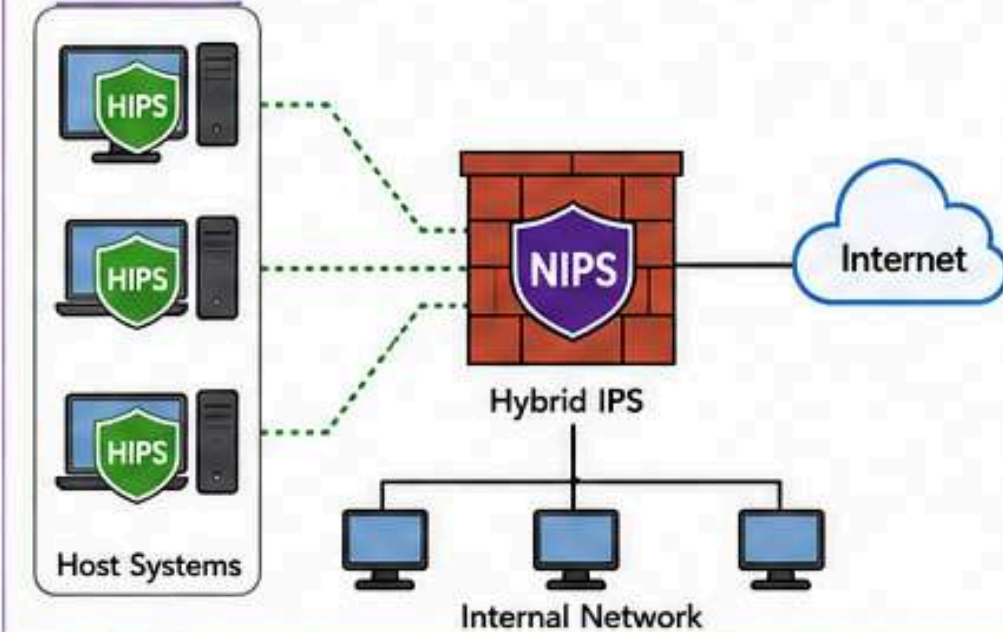
EXAMPLE

Protecting enterprise network

3) HYBRID IPS

Combination of Host-Based IPS and Network-Based IPS that provides security at both host and network levels.

DIAGRAM



FEATURES

- Combines HIPS and NIPS
- Provides layered security
- Detects both host and network attacks
- Offers better protection

ADVANTAGES

- Strong overall security
- Better attack detection
- Multi-layer protection

DISADVANTAGES

- Complex implementation
- High cost
- Requires skilled management

EXAMPLE

Large organizations, data centers, high-security environments

Operating System Security

- Operating System Security refers to the protection of an operating system from threats, attacks, unauthorized access, malware, and system misuse.
- It ensures confidentiality, integrity, and availability of system resources and data.

Objectives of Operating System Security

- Prevent unauthorized access
- Protect system files and resources
- Ensure data confidentiality
- Detect malicious activities
- Maintain system stability

Feature	Description
Authentication	Verifies user identity using passwords or biometrics
Access Control	Restricts unauthorized access to files and resources
Encryption	Protects sensitive data
Firewall	Blocks malicious network traffic
Antivirus Protection	Detects and removes malware
Audit Logs	Records user and system activities

Operating System Security

Advantages of Operating System Security

1. Protects sensitive information
2. Prevents unauthorized access
3. Detects malware attacks
4. Improves system reliability
5. Maintains data integrity

Limitations

- Requires regular updates
- Security misconfiguration can create vulnerabilities
- Cannot fully stop all cyber attacks

Types of Operating System Threats

- Virus - Malicious program that damages files
- Worm - Self-replicating malware spreading through networks
- Trojan Horse - Malicious software disguised as legitimate
- Spyware - Secretly collects user information
- Rootkits - Gain unauthorized administrator access

Application Security

- Application Security refers to the process of protecting software applications from threats, vulnerabilities, and unauthorized access during development and execution.
- It includes security measures to prevent attacks on applications and user data.

Objectives of Application Security

1. Protect application data
2. Prevent unauthorized access
3. Reduce software vulnerabilities
4. Ensure secure transactions
5. Maintain application reliability

Advantages of Application Security

1. Protects user data
2. Prevents cyber attacks
3. Improves application reliability
4. Secures online transactions
5. Reduces security vulnerabilities

Limitations

- Complex implementation
- Requires continuous updates
- Security testing can be costly

Application Security

Security Mechanisms in Application Security

Mechanism	Description
Authentication	Verifies user identity
Authorization	Controls user permissions
Encryption	Secures sensitive data
Input Validation	Prevents malicious input
Session Management	Protects user sessions
Error Handling	Prevents information leakage

Common Application Attacks

Attack	Description
SQL Injection	Malicious SQL queries attack database
Cross-Site Scripting (XSS)	Injects harmful scripts into webpages
Buffer Overflow	Overwrites memory causing crashes
Session Hijacking	Steals user session information
Brute Force Attack	Repeated login attempts

Security Maintenance

- Security Maintenance is the continuous process of monitoring, updating, and managing security measures to protect computer systems, networks, and applications from threats and vulnerabilities.
- It ensures that security systems remain effective and up to date.

Objectives of Security Maintenance

- Protect systems from new threats
- Maintain system reliability
- Reduce security vulnerabilities
- Ensure continuous protection
- Improve overall security performance

Activities in Security Maintenance

Activity	Description
Software Updates	Installing latest security patches
Antivirus Updates	Updating malware protection databases
Backup Management	Creating and maintaining backups
Log Monitoring	Monitoring system and network logs
Vulnerability Scanning	Identifying security weaknesses
Access Review	Checking user permissions regularly

Multilevel Security (MLS)

- Multilevel Security is a security model in which users and data are classified into different security levels, and access is granted according to those levels.
- It is mainly used in military and government systems.

Security Levels in MLS

Level	Description
Top Secret	Highest security level
Secret	High-level confidential data
Confidential	Moderate sensitive information
Unclassified	Public or low-risk data

Working of Multilevel Security

- Users are assigned security clearances.
- Data/resources are assigned classification levels.
- Access is granted only if user clearance matches security level.

Advantages of MLS

- 1.Strong data protection
- 2.Prevents unauthorized access
- 3.Protects confidential information
- 4.Suitable for government systems

Limitations

- Complex management
- Expensive implementation
- Difficult policy maintenance

Multilevel Security for Role-Based Access Control (RBAC)

Multilevel Security in RBAC combines security levels with role-based access control to provide secure and controlled access to resources.

Users get access according to both:

- Their role
- Their security clearance level

Working of MLS in RBAC

- Users are assigned roles.
- Roles are associated with security levels.
- Access is granted based on:
 - User role
 - Security clearance
- Unauthorized access is denied.

User Role	Security Level	Access Permission
Admin	Top Secret	Full access
Manager	Secret	Department files
Employee	Confidential	Limited data
Guest	Unclassified	Public information

Advantages

1. Better access control
2. High security
3. Organized permission management
4. Prevents privilege misuse

Limitations

- Complex role management
- Difficult implementation
- Requires continuous monitoring



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